Speaker Information

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Talk Information

Decoding the acoel nervous system: single-cell vs. single-nucleus RNA sequencing in *Praesagittifera naikaiensis*

Xenacoelomorpha — proposed either as sister to all other Bilateria or to Ambulacraria — represents one of the early branching bilaterian lineages, making it a valuable model for studying the ancestral organization of the nervous system. Acoela differ substantially from mammalian systems: they lack circulatory and vascular systems with body fluid in direct contact with cells, yet possess well-developed nervous system. Here we compare scRNA-seq and snRNA-seq in the acoel *Praesagittifera naikaiensis*. Both methods yielded comparable data quality. snRNA-seq showed enrichment of neuronal, muscle, and ciliated cell transcripts, and revealed additional cell populations not detected by scRNA-seq, including neural clusters. In addition, we identified genes encoding neuropeptides, glutamate decarboxylase (*GAD*), and the transcription factor *Six3*, with expression enriched in the anterior neural region. These findings provide a starting point for exploring the origins of anterior central nervous system patterning in Bilateria.

Research Overview (for collaboration discussion)

[Research Interests] Evolution of the nervous system in early-branching animals

[Current Research Projects]

- Cell type diversity in acoels, with a focus on neural populations and their development/differentiation
- Biochemical innovation and function of the monoaminergic system in metazoa (joint project with University of Perugia, Italy)

[Techniques] snRNA-seq, scRNA-seq, Visium HD, in situ HCR, RNAi knockdown in acoels

[Aspects or techniques of interest for future collaboration] Cross-species transcriptomic comparisons across Xenacoelomorpha (and Bilateria); **establishing the calcium imaging or electrophysiological approaches** for detecting neural activity **in living acoels**

Short Bio

Ryo Nakamura is a researcher at the Ushimado Marine Institute, Okayama University, on the shores of the Seto Inland Sea — home to the acoel *Praesagittifera naikaiensis*. Research in the lab investigates how the nervous system evolved in Bilateria, using this locally abundant yet phylogenetically interesting animal as a model organism.